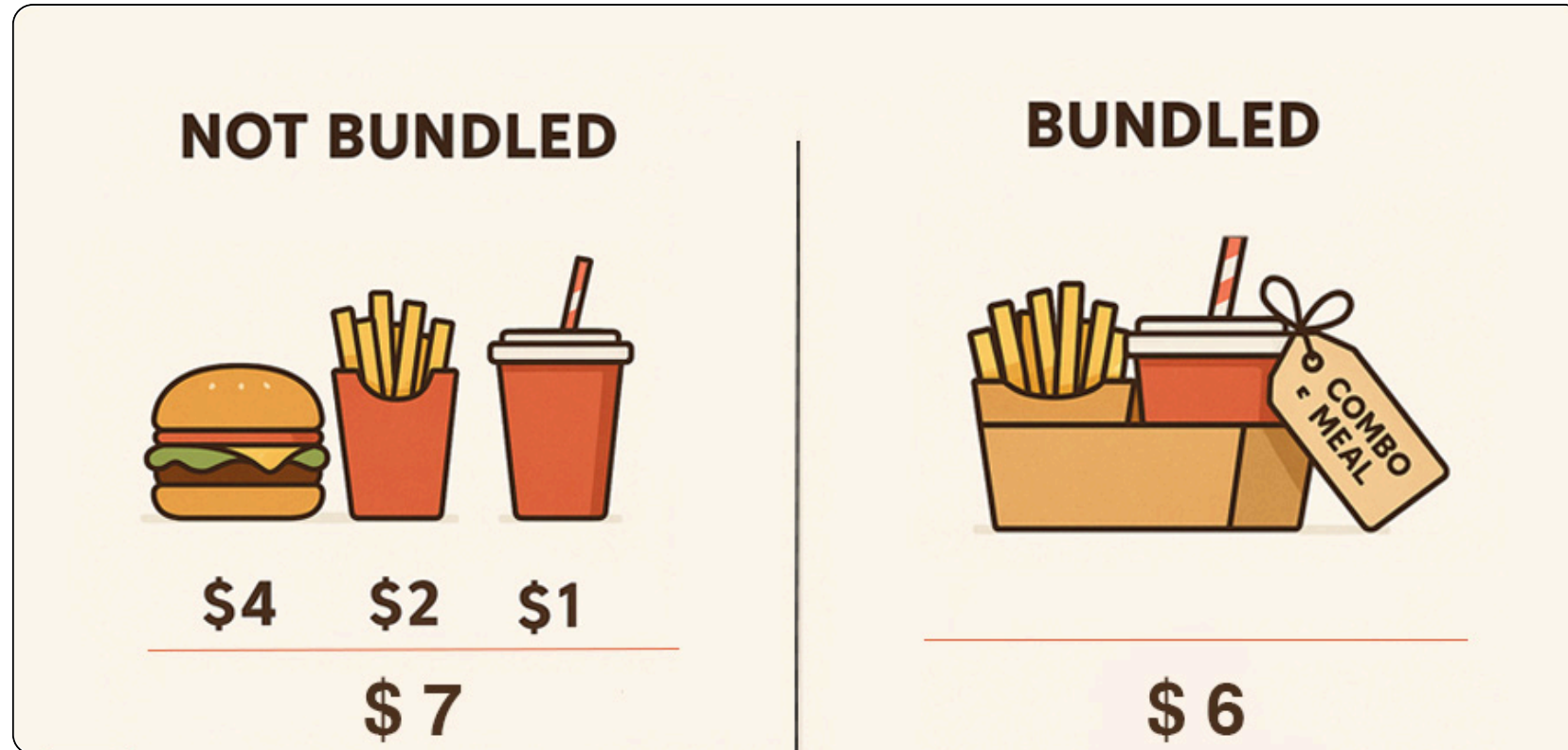


Association Rules

“What goes with What?”

Market-basket Analysis/Affinity Analysis



Main Purpose

Cross-Selling/UpSelling

Association Rules

"Market-based Analysis" / Affinity Analysis

I] Generate frequent item-set $\Rightarrow$ 'Apriori Algorithm' / FP growth.

Generating frequent item sets: The Apriori Algorithm

For k products...

1. Set minimum support criterion
2. Generate list of one-item sets that meet the support criterion
3. Use list of one-item sets to generate list of two-item sets that meet support criterion
4. Use list of two-item sets to generate list of three-item sets that meet support criterion
5. Continue up through k -item sets

① Support = 0.50

$$\text{Support} = \frac{\text{How many transactions are involved in 1 product / 'n' product}}{\text{Tot. No. of Transaction.}}$$

② P_1

P_2

P_3

P_4

P_5

P_7

P_8

$$P_1 = \frac{5}{10} = 0.50$$

$$P_2 = \frac{6}{10} = 0.60 \Rightarrow P_1, P_2, P_3, P_4, P_5.$$

$$P_3 = \frac{5}{10} = 0.50$$

...

$$\textcircled{3} P_1, P_2 - \frac{3}{10} = 0.30 \quad \textcircled{\checkmark}$$

$$P_1, P_3 - \frac{3}{10} = 0.30$$

$P_1, P_4 \quad \textcircled{\checkmark}$

$$\Rightarrow P_1, P_2 \quad \textcircled{\checkmark}$$

P_1, P_5

$$\Rightarrow P_1, P_4$$

$P_2, P_3 \quad \textcircled{\checkmark}$

$$\Rightarrow P_2, P_5$$

P_2, P_4

$P_2, P_5 \quad \textcircled{\checkmark}$

$$\Rightarrow P_2, P_5$$

P_3, P_4

$P_3, P_5 \quad \textcircled{\checkmark}$

$$\Rightarrow P_3, P_5$$

P_4, P_5

$$\textcircled{4} P_1, P_2, P_3 \quad \textcircled{\checkmark}$$

P_2, P_3, P_4

$$P_1, P_2, P_5 \quad \textcircled{\checkmark}$$

P_2, P_3, P_5

$$P_1, P_2, P_4 \quad \textcircled{\checkmark}$$

⑤

$$\text{Confidence} = \frac{\text{Transactions are involved 1 product / 'n' product}}{\text{Tot. no. of antecedent.}}$$

$$P_1 = \frac{5}{5} = 1.0 \quad \textcircled{\checkmark}$$

$$P_2 = \frac{6}{6} = 1.0 \quad \textcircled{\checkmark}$$

...

$$P_8 = 1.0 \quad \textcircled{\checkmark}$$

$$\Rightarrow P_1, P_2 = \frac{3}{5} = 0.6 \quad \textcircled{\checkmark}$$

(P_1, P_2)

Confidence = 0.6

What Is Association Rule Mining?

- Rule form

Antecedent $\rightarrow$ Consequent [**support**, **confidence**] And **lift**

(support and confidence are user defined measures of interestingness)

- Examples

- $\text{buys}(x, \text{"computer"}) \rightarrow \text{buys}(x, \text{"financial management software"})$
[0.5%, 60%]
- $\text{age}(x, \text{"30..39"}) \wedge \text{income}(x, \text{"42..48K"}) \rightarrow \text{buys}(x, \text{"car"})$
[1%, 75%]